Thomas Janssoone

Research Scientist – Speech, Multimodal AI & LLM Evaluation

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Research Profile

My research asks one question across settings: how to extract a reliable signal about a hidden state (affect, rapport, patient risk, genome function, agent coordination) from multimodal or sequential data, and how to make the resulting model trustworthy. I approach it measurement-first: interpretable models (SHAP, LIME, association rules), pre-registered ablations, and confidence intervals before conclusions. Over 15+ years I have applied it to social signal processing (PhD in affective computing, Sorbonne Université), clinical prediction on national health data, real-time voice interaction for social robots and emergency-call triage, long-context genomic foundation models, and multi-agent LLM systems. **Current interests:** speech and multimodal affective AI, evaluation of LLMs and voice agents, long-context foundation models, internal-state control of LLM agents.

Research and Professional Experience

AI Researcher, Plant Genomic Foundation Models, Living Models, Paris — Apr 2026 – Sep 2026

- Co-developed BOTANIC-1, plant-specialist genomic language models (bidirectional Mamba2, 318M to 3.2B parameters) pre-trained to a 128 kbp context, twice the previous plant-model record, on a 320-genome corpus spanning the land plants. Weights and pre-training data released on Hugging Face; co-author of the bioRxiv preprint.
- Ran pre-training on a 10-node, 120-GPU SLURM cluster (University of Florida), orchestrating training runs with LLM agents.
- Built the agentic use case validating BOTANIC-1 as a research tool: a local Gemma agent that keeps proprietary breeding genotypes on-premises and calls BOTANIC-1 as a zero-shot variant-effect scorer through an internal bioinformatics tool suite.
- Led the NeurIPS 2026 Competition Track submission for PlantGenomeBench, a unified evaluation framework with four tracks (molecular signals, cross-species generalization, phenotype-linked prediction, multimodal integration), co-organised with the University of Florida.
- Exploratory work on long-context architecture variants (memory caching for DNA language models) and gene-by-environment (G×E) modelling.

AI Researcher, Medical Speech Analysis, e-sensia, Paris — Sep 2025 – Feb 2026

- Architected low-latency, real-time audio streaming pipelines (WebSockets, WebRTC) for emergency medical triage on live calls.
- Built real-time inference loops with Whisper, Wav2Vec and EnCodec to extract vocal biomarkers and social signals within latency budgets; evaluated MedGemma for medical use cases.
- Wrote the ESYNAPSE grant, selected as a France 2030 "Pionniers de l'IA" laureate (up to €10M); prepared Research Tax Credit (CIR) applications and competitive funding proposals.

Multimodal Machine Learning Expert, Social Robotics, Enchanted Tools, Paris — Apr 2022 – Aug 2025

- Engineered the real-time voice interaction stack of the Mirokai social robot (STT, LLM/VLM dialogue management, TTS) with sub-second loops, voice activity detection, turn detection and barge-in, with a focus on affective interaction.
- Deployed WebSockets/WebRTC transport for bidirectional audio and video between embedded hardware (NVIDIA Jetson) and cloud or local GPU backends.
- Benchmarked and fine-tuned LLMs and SLMs (prompt engineering, fine-tuning, RAG, tool calling for robot actions) on real and synthetic data generated in-house to anticipate real-world use cases and bias, with daily metrics (BLEU, function-calling accuracy, LLM-as-a-judge).
- Fine-tuned, voice-cloned and benchmarked TTS models (Parler-TTS, Coqui, Matcha-TTS) in English, French and Japanese on Jetson Orin NX/AGX, trading off speed, expressivity and tone fidelity.
- Evaluated third-party versus open-source solutions and took part in hardware integration and industrialisation.
- Research: co-supervised a PhD on human-robot interaction in healthcare facilities (IEEE ICCMI 2025); co-organised the ACE workshops at ICMI 2023 and IVA 2024.

Postdoctoral Researcher, Interpretable Social AI, Inria (COML, J. Cassell), Paris — Apr 2021 – Dec 2021

- Research on conversational rapport for virtual agents that must reach both a rapport level and a task goal (for example in peer tutoring).
- Drove the research direction of an interpretable, dyadic multimodal rapport estimator in videoconference tutoring, analysed with SHAP (HHAI 2022).

- Built a Python library for multimodal social-signal extraction and an open-source AI-assisted annotation tool for children's communication patterns.
- Explored generative conversational strategies (RL-GAN, BlenderBot, GPT-2). Best Reviewer award, ICMI 2021.

Research Scientist, Healthcare Machine Learning, Sêmeia, Paris Apr 2018 – Jan 2021

- Led a team of 2 data scientists working on SNIIRAM, the French national health database (66 million people: diagnoses, hospitalisations, drug purchases, context).
- Combined weak signals (pharmacy purchases, care context) to identify over 90% of patients at risk of early treatment discontinuation.
- Built a Python library of statistical, survival-analysis, sequence-mining and deep-learning models, with SHAP/LIME explanations for medical staff; dimensionality reduction (UMAP) on BERT embeddings; large-scale aggregates with SQL, pandas and Polars.
- Contributed to the design of a secure server for health data and to proposals for the Health Data Hub; defined the research roadmap and ran the scientific watch on interpretable ML.
- Published at the ML4H and AI for Social Good workshops (NeurIPS 2018), PRIME (MICCAI 2018) and CNIA 2019.

Founder, Smart Farming System, Paris Jun 2018 – Nov 2022

- Founded a start-up designing autonomous vertical urban farms controlled by AI; first product, pop-farm, grows herbs and fruits inside supermarkets, without pesticides or transport.

Visiting Research Scholar (PhD), USC Institute for Creative Technologies, Los Angeles Jul 2017 – Sep 2017

- With Prof. Stefan Scherer: deep-learning models of emotionally coloured dyadic interaction, using domain separation networks.

R&D Engineer, Human-Computer Interaction, LIG, Human-Computer Interaction group, Grenoble Feb 2013 – Jul 2014

- Designed and evaluated guidance for 3D gestures to control lighting (screen, augmented reality on the hand, mirror), with Kinect tracking and Ogre3D rendering, within an industrial FUI project (AVI 2016).

R&D Engineer, Computer-Assisted Surgery, TIMC-IMAG (CAMI team) with Imactis and SurgiQual Institute, Grenoble Jan 2010 – Apr 2012

- Built a CT-guided navigation prototype (specifications, risk analysis, impact of breathing on organ motion) and software components for medical navigation at Grenoble University Hospital.

Independent Research (2026)

- **Hormonal Agents: what hormone-inspired controllers actually do.** LLM agents with six hormone-like internal variables evolving by differential equations. Pre-registered ablations across the iterated prisoner's dilemma, a coding task and τ^2-bench, on local models from 1B to 26B parameters, showed that the gains came from the state description in the prompt, not from the dynamics (frozen state: 99.8% cooperation versus 99.7%). Moving the state into the model's activations, a per-partner state is the one mechanism that beats replay controls and targets a betraying partner. Paper in preparation.
- **Concilium.** A council of small language models that deliberate and vote. First testbed: one model fine-tuned per member of the French Assemblée nationale to predict votes and speeches; extended to the Conseil de Paris, with activation steering of tone and expressive speech synthesis of generated speeches. Paper in preparation.
- **Local ASR and diarization benchmark: where words get lost.** 11 ASR systems (Whisper, Parakeet, Qwen3-ASR, Voxtral, Kyutai, Canary, Nemotron) and 3 diarizers on 162 English and French datasets of synthetic noisy multi-speaker scenes, run on Apple Silicon. Error analysis beyond WER (cpWER, DER, overlap, backchannels) with paired hierarchical bootstrap confidence intervals. Main finding: speaker attribution, not transcription, causes 38–63% of errors. Found and fixed a streaming bug in the transcribe.cpp multi-talker port (WER 54.9% to 31.7%); also tested full-duplex models (Moshi, PersonaPlex) on turn-taking probes.

Education

PhD in Computer Science, Affective Computing, Sorbonne Université (ISIR) and Télécom Paris Oct 2014 – Jan 2018

- Machine learning for temporal information in social interaction, for both recognition and synthesis.
- Designed a sequence-mining method that infers stance-specific temporal association rules from visual (facial expressions, head movements) and audio (prosody, turn-taking) features, used to animate embodied conversational agents expressing dominance or friendliness. Validated against the humanities literature and through perceptive studies, using text-to-speech, voice morphing and facial animation.

Engineering degree (MSc), Applied Mathematics and Computer Science, Ensimag, Grenoble INP Sep 2006 – Jul 2010

- Specialisation in image and virtual reality; probabilistic machine learning (EM, Gaussian mixture models, HMMs, Kalman filters).
- Exchange semester at Eindhoven University of Technology (expert systems, medical image processing), Sep 2009 – Jan 2010.
- Final internship, TIMC-IMAG (CAMI team): a learning environment for ultrasound-guided prostate biopsies, with user-interface design and evaluation (MMVR 2011).

Academic Service and Awards

- Co-organiser, ACE workshop (artificial character embodiment in multimodal interaction): ICMI 2023 and IVA 2024.
- Co-organiser, PlantGenomeBench, NeurIPS 2026 Competition Track submission.
- Program committee: IVA 2023, IVA 2024, ECAI 2024.
- Best Reviewer award, International Conference on Multimodal Interaction (ICMI) 2021.

Supervision, Funding and Outreach

- **Supervision:** co-supervised a PhD in human-robot interaction (Enchanted Tools); managed a team of 2 data scientists (Sêmeia).
- **Funding:** lead writer of the ESYNAPSE proposal, France 2030 "Pionniers de l'IA" laureate (up to €10M); Research Tax Credit (CIR) applications; proposals to the French Health Data Hub.
- **Outreach:** science mediation at the Cité des Sciences et de l'Industrie, Paris, during the PhD.
- **Training:** International Summer School in Affective Sciences (ISSAS); Responsible Conduct of Research (CITI, USC).

Technical Skills

- **Speech & audio:** ASR, speaker diarization, TTS and voice cloning, VAD, turn-taking, full-duplex models, real-time streaming (WebSockets, WebRTC)
- **LLMs & agents:** evaluation (LLM-as-a-judge, function calling), fine-tuning, RAG, tool calling, activation steering, multi-agent systems, on-premises deployment
- **Machine learning:** deep learning, multimodal fusion, foundation-model pre-training (long context, Mamba/SSM), sequence mining, survival analysis, interpretability (SHAP, LIME), statistical evaluation (bootstrap, pre-registration)
- **Engineering:** Python, PyTorch, Hugging Face, NVIDIA NeMo, scikit-learn, pandas, Polars, NumPy, SQL, C++, ONNX, Docker, GitLab CI, SLURM, MLflow, Weights & Biases, DVC, Linux
- **Languages:** French (native), English (fluent, C1/C2)

Publications

1. A. Barozet, V. Cabeli, J. Ogier du Terrail, A. Rukhovich, **T. Janssoone**, G. Klajer, Z. Sheikhitarghi, G. Andrews, C. Veran, L. Strouk. BOTANIC-1: a series of long-context plant genomic foundation models in the agentic era. *bioRxiv*, 2026.
2. R. Vallée, V. Aubergé, E. Chevalier, L. Prégaldiny, S. Dacunha, **T. Janssoone**. Social logistics robot in healthcare facilities: perceptive power of presenting the robot as a subject versus an object with expressive variation. *IEEE 4th International Conference on Computing and Machine Intelligence (ICCMI)*, 2025.
3. E. Ceccaldi, B. Biancardi, S. Falcone, S. Ferrando, G. Gorisse, **T. Janssoone**, A. Martin Coesel, P. Raimbaud. ACE: how artificial character embodiment shapes user behaviour in multi-modal interaction. *25th International Conference on Multimodal Interaction (ICMI)*, 2023.
4. O. Goumri, **T. Janssoone**, et al. Automatic detection of gaze and smile in children's video calls. *International Conference on Multimodal Interaction (ICMI)*, 2023.
5. G. Grimberg, **T. Janssoone**, C. Clavel, J. Cassell. Estimating rapport in conversations: an interpretable and dyadic multi-modal approach. *First International Conference on Hybrid Human-Artificial Intelligence (HHAI)*, 2022.
6. **T. Janssoone**, C. Clavel, K. Bailly, G. Richard. The POTUS corpus, a database of weekly addresses for the study of stance in politics and virtual agents. *Language Resources and Evaluation Conference (LREC)*, 2020.
7. **T. Janssoone**, P. Rinder, C. Bic, D. Kanoun, P. Hornus. Modèles d'apprentissage automatique de la persistance aux médicaments : application au cancer du sein. *Conférence Nationale en Intelligence Artificielle (CNIA)*, 2019.
8. **T. Janssoone**, C. Bic, D. Kanoun, P. Hornus, P. Rinder. AI for patient support: predictive model of medication non-adherence. *AI for Social Good Workshop, NeurIPS*, 2018.

9. **T. Janssoone**, C. Bic, D. Kanoun, P. Hornus, P. Rinder. Machine learning on electronic health records: models and features usages to predict medication non-adherence. *Machine Learning for Health (ML4H) Workshop, NeurIPS*, 2018. arXiv:1811.12234.
10. **T. Janssoone**, P. Rinder, P. Hornus, D. Kanoun. Predictive patient care: survival model to prevent medication non-adherence. *PRIME Workshop, MICCAI*, 2018.
11. **T. Janssoone**, C. Clavel, K. Bailly, G. Richard. SMART : règles d'associations temporelles de signaux sociaux pour la synthèse d'un agent conversationnel animé avec une attitude spécifique. *Revue d'Intelligence Artificielle*, 2017.
12. W. Delamare, **T. Janssoone**, C. Coutrix, L. Nigay. Designing 3D gesture guidance: visual feedback and feedforward design options. *International Working Conference on Advanced Visual Interfaces (AVI)*, 2016.
13. **T. Janssoone**, C. Clavel, K. Bailly, G. Richard. Using temporal association rules for the synthesis of embodied conversational agents with a specific stance. *International Conference on Intelligent Virtual Agents (IVA)*, 2016.
14. **T. Janssoone**, C. Clavel, K. Bailly, G. Richard. Des signaux sociaux aux attitudes : de l'utilisation des règles d'association temporelle. *WACAI, Workshop Affect, Compagnon Artificiel, Interaction*, 2016.
15. **T. Janssoone**. Temporal association rules for modelling multimodal social signals. *ACM International Conference on Multimodal Interaction (ICMI)*, 2015.
16. **T. Janssoone**, G. Chevreau, L. Vadcard, P. Mozer, J. Troccaz. Biopsym: a learning environment for trans-rectal ultrasound guided prostate biopsies. *Medicine Meets Virtual Reality (MMVR)*, 2011.